Regular Elements as Twisted Principal Series

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1 Centralizers

Let G and H be reductive groups over a local field.

In Langlands, *functoriality* is the principle that a map ${}^L H \to {}^L G$ induces a *transfer*

$$\left\{ \begin{array}{c} \text{admissible irreps} \\ \text{of } H \end{array} \right\} \to \left\{ \begin{array}{c} \text{admissible irreps} \\ \text{of } G \end{array} \right\}.$$

One of the basic testing grounds was and is *endoscopy*.

Roughly, H is *endoscopic* to G iff ${}^L H = C_{{}^L G}(s)^\circ$ for some semisimple element $s \in {}^L G$.

Today's story is **not** an analogue of endoscopy for Weyl groups, but...

Let W be a well-generated (complex) reflection group.

An element of its braid group

$$Br_W := \langle (\sigma_w)_{w \in W} \mid \sigma_w \sigma_s = \sigma_{ws} \text{ for } ws > w \rangle$$

is *periodic* iff it has a central power.

(Springer '74 + Broué–Michel '97)

An element $w \in W$ is *regular* iff σ_w is periodic.

In this case, $C_W(w)$ is also a reflection group.

Example The identity e.

Example The longest element $w_\circ$.

Example For irreducible W, Coxeter elements.

Now take W to be the Weyl group of a finite reductive group

$$G = \mathbf{G}^F, \quad \text{where } F \text{ is a } q\text{-Frobenius map.}$$

A choice of F-stable, maximally F-split $\mathbf{A} \subseteq \mathbf{G}$ yields:

$$\begin{aligned} \{\text{maximal tori of } G\}/\text{conj.} &\simeq W/\text{conj.} \\ T &= \mathbf{A}^{wF} \end{aligned}$$

(Deligne–Lusztig '76)

All irreps of G arise from such T via *cohomological* induction maps

$$R_T^G \colon K_0(T) \to K_0(G).$$

Special features when T is of type $[w]$ for *regular* w.

An irrep of G is *unipotent* iff it belongs to

$$\mathrm{Uch}_T^G = \{\rho \mid (\rho, R_T^G(1))_G \neq 0\} \quad \text{for some } T.$$

For such ρ, a *generic degree* $\mathrm{Deg}_\rho(x) \in \mathbf{Q}[x]$ such that

$$\boxed{\deg \rho = \mathrm{Deg}_\rho(q).}$$

(Lusztig, Broué–Malle–Michel '93)

If w is regular of order m, and T is of type $[w]$, then

$$\mathrm{Uch}_T^G = \{\text{unipotent } \rho \mid \mathrm{Deg}_\rho(\zeta_m) \neq 0\}.$$

And $\exists\, \chi_T^G \colon \mathrm{Uch}_T^G \xrightarrow{\sim} \mathrm{Irr}(C_W(w))$ such that

$$\boxed{\deg \chi(\rho) = |\mathrm{Deg}_\rho(\zeta_m)|.}$$

Proved case by case. Morally, why the bijection?

Conjecture (Broué–Malle '93) $H_T^G(x)$ such that

$$\boxed{\begin{aligned} H_T^G(q) &\simeq \mathrm{End}_G(\mathrm{H}^*_{\mathrm{c,\acute{e}t}}(X_w)),\\ H_T^G(\zeta_m) &\simeq \mathbf{C}C_W(w). \end{aligned}}$$

χ_T^G should arise from a double centralizer theorem.

Example Write $A := \mathbf{A}^F$. It is of type $[e]$.

$$X_e = (\mathbf{G}/\mathbf{B})^F = G/B.$$

$R_A^G(1)$ consists of the *unipotent principal series*: the irreps occuring in $\mathbf{C}[G/B]$.

$H_A^G(x)$ is the usual Hecke algebra of W.

Example Take $G = \mathrm{Sp}_4(\mathbf{F}_q)$ and T of *Coxeter type*.

Any Coxeter element c has order $h = 4$.

	$\mathrm{Deg}_\rho(x)$	$\mathrm{Deg}_\rho(1)$	$\mathrm{Deg}_\rho(\zeta_4)$
	1	1	1
	$\frac{1}{2}x(x+1)$	1	$\cdot$
	$\frac{1}{2}x(x+1)^2$	2	-1
	$\frac{1}{2}x(x+1)$	1	$\cdot$
	x^4	1	1
θ_{10} $\rightarrow$	$\frac{1}{2}x(x-1)^2$	$\cdot$	1

$W = C_W(e)$ is dihedral of order 8, while $C_W(c)$ is cyclic of order 4.

Anything special about $\mathrm{Uch}_A^G \cap \mathrm{Uch}_T^G$?

Let V be the reflection representation of W.

($\approx$ Lusztig '78) For G almost simple and $\rho \in \mathrm{Uch}_A^G$,

$$\mathrm{Deg}_\rho(\zeta_h) = \begin{cases} (-1)^k & \chi_A^G(\rho) = \Lambda^k(V), \\ 0 & \text{else.} \end{cases}$$

So for, T of Coxeter type, χ_A^G restricts to

$$\mathrm{Uch}_A^G \cap \mathrm{Uch}_T^G \xrightarrow{\sim} \{\Lambda^k(V) \mid 0 \le k \le \mathrm{rk}(G)\}.$$

For $W = S_n$, indirectly recovers Frobenius's result:

$$\chi(n\text{-cycle}) = \begin{cases} (-1)^k & \chi = \Lambda^k(V), \\ 0 & \text{else.} \end{cases}$$

2 Combinatorics

$\mathrm{Uch}_A^G \cap \mathrm{Uch}_T^G$ is implicit throughout my work with Nathan.

For any word $\vec{s}$ in simple reflections, let

$$\mathfrak{D}(\vec{s}) = \{\text{Deodhar walks for } \vec{s}\}.$$

These are walks $\vec{w}$ in W, starting and ending at e, such that for all i:

- either $w_{i+1} = w_i s_{i+1}$ and $w_{i+1} = w_i$.
- $w_{i+1} \leq w_i s_{i+1}$. *If you can descend, you must.*

Let $d_{\vec{w}}$, *resp.* $e_{\vec{w}}$, count the descents, *resp.* stays.

These appear in point counts of Deodhar cells.

(Galashin–Lam–T.–Williams '22)

For any *Coxeter word* $\vec{c}$, and $p > 0$ coprime to h,

$$\frac{1}{q^{rp/2}(1-q)^r} \sum_{\vec{w} \in \mathfrak{D}(\vec{c}^p)} q^{d_{\vec{w}}} (q-1)^{e_{\vec{w}}}$$

$$= \sum_{\chi \in \mathrm{Irr}(W)} \mathrm{Deg}_\chi(\zeta_h)(1, [\Delta_{p/h}(\chi)]_q)_W$$

$$= (1, [L_{p/h}(1)]_q)_W \quad \leftarrow \ q\text{-Catalan}$$

where:

- r is the rank of W.
- $\mathrm{Deg}_\chi = \mathrm{Deg}_\rho$ when $\chi_A^G(\rho) = \chi$.
- Δ, *resp.* L, denotes Verma, *resp.* simple, rational Cherednik modules.
- $[M]_q = \sum_i \mathrm{gr}_i\, q^{i/2} M_i \in K_0(W)((q))$.

In general:

Δ's have easy characters, but are infinite-dimensional.

L's are finite-dimensional, but their characters are mysterious.

(Berest–Etingof–Ginzburg, Chmutova–Etingof '03)

$$\sum_k (-1)^k [\Delta_{p/h}(\Lambda^k(V))]_q = [L_{p/h}(1)]_q.$$

Recall that $\mathrm{Deg}_\chi(\zeta_h) = \begin{cases} (-1)^k & \chi_A^G(\rho) = \Lambda^k(V), \\ 0 & \text{else.} \end{cases}$

If p/h simplifies to d/m in lowest terms:

How does $\sum_\chi \mathrm{Deg}_\chi(\zeta_m)[\Delta_{d/m}(\chi)]_q$ expand into L's?

3 Cohomology

Let $\hat{G}$ be the associated form over $\mathbf{C}(\!(z)\!)$. Again,

$$\{\text{maximal tori of } \hat{G}\}/\text{conj.} \quad \simeq \quad W/\text{conj.}$$

$\gamma \in \mathrm{Lie}\,\hat{G}$ is *homogeneous* of slope d/m iff

$$\gamma(c^m z) \underset{\text{conj.}}{\sim} c^d \gamma(z) \quad \text{for } c \in \mathbf{C}^\times.$$

For such γ generic* enough, $C_{\hat{G}}(\gamma)$ is a maximal torus whose type is regular of order m.

Example Take $\hat{G} = \mathrm{SL}_2(\mathbf{C}(\!(z)\!))$:

$\begin{pmatrix} z^d & \\ & -z^d \end{pmatrix}$ has slope d; $\begin{pmatrix} & 1 \\ z^d & \end{pmatrix}$ has slope $d/2$.

* regular semisimple

For generic homogeneous γ of slope d/m, the

$$\text{affine Springer fiber } \mathcal{F}l_\gamma \subseteq \mathcal{F}l_{\check{G}}$$

has a $\mathbf{C}^\times$-action studied by Lusztig–Smelt, Fan, Sage, Sommers, Varagnolo–Vasserot, Oblomkov–Yun, ...

Let $T \subseteq G$ have the same type as $C_{\check{G}}(\gamma)$, and let

$$E_\gamma = \mathrm{gr}^{\mathsf{P}}_* \, \mathrm{H}^*_{\mathbf{G}_m,\mathrm{loc}}(\mathcal{F}l_\gamma)_{\mathrm{st}}.$$

(Oblomkov–Yun) Rational Cherednik action* on E_γ.

Conjecture (T.–Xue '23) $\mathrm{End}_{\mathrm{RCA}}(V_{d/m}) \simeq H_T^G(1)$,

$$[E_\gamma]_q = \sum_{\rho \in \mathrm{Uch}_A^G \cap \mathrm{Uch}_T^G} [\Delta_{d/m}(\chi_A^G(\rho))]_q \otimes \chi_T^G(\rho)|_{x\to 1}.$$

* assuming γ elliptic

Theorem (T.–Xue '23) The conjecture holds when:

- m is the (twisted) Coxeter number of $\check{G}$.
- $(\check{G}, m) = ({}^2A_2, 2), (C_2, 2), (G_2, 3), (G_2, 2)$.

The mysterious analogy:

$\mathbf{F}_q$	(q, q)	$\mathrm{H}^*_c(X_e) \otimes_G \mathrm{H}^*_c(X_w)$
$\mathbf{C}(\!(z)\!)$	$(\zeta_m, 1)$	E_γ

Out of time to explain how E_γ is *level-rank duality* between $H_A^G(\zeta_m)$ and $H_T^G(1)$.

Thank you for listening.